

```

M:=9;
P1:=5;
P2:=10;
P3:=20;
P4:=30;
LONG:=26;
SHORT:=12;
MA1:= EMA(C,P1);
MA2:= EMA(C,P2);
MA3:= EMA(C,P3);
MA4:= EMA(C,P4);
MA5:= MA(C,120);
换手率:=(VOL/CAPITAL) ;

{===== 引用布林线 =====}
N:=26; P:=20;
MID:= MA(CLOSE,N);
UPPER:=MID + P/10*STD(CLOSE,N);
LOWER:=MID - P/10*STD(CLOSE,N);
多头排列:=MA1>MA2 and MA2>MA3 and MA2>MA4 and MA4>ref(MA4,5);
空头排列:=MA1<MA2 and MA2<MA3 and MA3<MA4;
上升趋势:=MA4>ref(MA4,3) and MA4>ref(MA4,1) and MA1>MA4;
下跌趋势:=MA4<ref(MA4,3) and MA4<ref(MA4,1) and MA1<MA4;
{
EXPLAIN(C>0,'===== 【综合解盘】 ====='),COLORRED; EXPLAIN(C>0,'注意：目前解盘内容仅适用'); EXPLAIN(C>0,'于个股的日线周期分析参考');
}
{===== 【基本信息提示】 =====}
EXPLAIN(C>0,'—— 【基本信息提示】 ——'),COLORYELLOW;
EXPLAINEX(1,'流通盘(万股): ',FINANCE(3), 1,0);
EXPLAINEX(1,'总股本(万股): ',FINANCE(0), 2,0);
EXPLAINEX(1,'每股收益(元): ',FINANCE(29),3,0);
{
EXPLAINEX(1,'动态市盈率: ',close/FINANCE(29),2,0);
}
EXPLAINEX(close>FINANCE(34),'每股净资产(元): ',FINANCE(31),3,0) ;
EXPLAINEX(close<FINANCE(34),'每股净资产(元): ',FINANCE(31),3,0) ;
EXPLAINEX(1,'净资产收益率 %: ',FINANCE(33),2,0);
{EXPLAIN(C>0,'—————'),COLOR666666;}
EXPLAINEX(1=1,'获利盘比例值 %: ',WINNER(c)*100,2,0);
{
EXPLAINEX(1,'总股本:',FINANCE(1),'流通股:',FINANCE(7),2,0);
EXPLAINEX(1,'主营利润: ',FINANCE(21),2,0);
EXPLAINEX(1,'利润增长: ',FINANCE(54),2,0);
EXPLAIN(1,F10TEXT(F10FIND('发行价',1),30));
EXPLAINEX(1,'获利盘 %: ',winner(close)*100,2,0),COLORGREEN;
}
{
EXPLAIN(C>0,'—— 【板块信息提示】 ——'),COLORBLUE;
EXPLAIN(1,BLKNAME);
EXPLAIN(1,STKNAME+STKLABEL+'\\n所属板块:' +BLKNAME)
}
{===== 【MACD指标分析】 =====}
EXPLAIN(C>0,'—— 【经典指标分析】 ——'),COLORYELLOW;
DIF:EMA(CLOSE,SHORT)-EMA(CLOSE,LONG),COLORCCFFFF;
DEA:EMA(DIF,M),COLOR58FF58;
MACD:=2*(DIF-DEA),COLORFFF00,LINETHICK0;
STICKLINE(MACD>=0 AND MACD<=REF(MACD,1),0,MACD,6,0),COLOR6058FF;
STICKLINE(MACD>=0 AND MACD< REF(MACD,1),0,MACD,6,1),COLOR6058FF;
STICKLINE(MACD<0 AND MACD>=REF(MACD,1),0,MACD,6,1),COLORFFF00;
STICKLINE(MACD<0 AND MACD< REF(MACD,1),0,MACD,6,0),COLORFFF00;

```

```

多头趋势:=IF(DIF>=0 AND DEA>=0 OR (DIF<0 AND DEA>=0),1,0);
空头趋势:=DIF<0 AND DEA<0 OR (DIF>=0 AND DEA<0);
MACD绿柱缩短:=MACD<0 AND ABS(MACD)< REF(ABS(MACD),1) and ref(MACD,1)<0;
MACD红柱缩短:=MACD>0 AND MACD< REF(MACD,1);
缩头:=IF(MACD>0 AND REF(MACD,1)>REF(MACD,2) AND REF(MACD,2)>REF(MACD,3) AND MACD< REF(MACD,1),1,0) and ref(MACD,1)>0.150;
缩脚:=IF(MACD<0 AND REF(MACD,1)< REF(MACD,2) AND REF(MACD,2)< REF(MACD,3) AND MACD>REF(MACD,1),1,0) and ref(MACD,1)<-0.100;
金叉:=CROSS(DIF,DEA);
死叉:=CROSS(DEA,DIF);
0轴下死叉:=死叉 AND DEA<0;
0轴上死叉:=死叉 AND DEA>0;
0轴下金叉:=金叉 AND DEA<0;
0轴上金叉:=金叉 AND DEA>0;
0轴上二次金叉:=COUNT(CROSS(DIF,DEA) AND DEA>0,20)=2 AND CROSS(DIF,DEA) AND DEA>0;
0轴上三次金叉:=COUNT(CROSS(DIF,DEA) AND DEA>0,20)=3 AND CROSS(DIF,DEA) AND DEA>0;
{-----}
{
EXPLAIN(缩头, '注意MACD柱状缩头');
EXPLAIN(缩脚, '注意MACD柱状缩脚');
EXPLAIN(MACD<0 AND ABS(MACD)>REF(ABS(MACD),1), 'MACD绿柱增长');
EXPLAIN(MACD绿柱缩短 and not(缩脚), 'MACD绿柱缩短');
EXPLAIN(MACD>0 AND MACD>REF(MACD,1), 'MACD红柱增长');
EXPLAIN(MACD红柱缩短 and not(缩头), 'MACD红柱缩短');
}
EXPLAIN(金叉 and not(0轴下金叉) and not(0轴上金叉) and not(0轴上二次金叉), '金叉--属强势信号');
EXPLAIN( 0轴下死叉, 'DEA和DIF在0轴下死叉');
EXPLAIN( 0轴下金叉, 'DEA和DIF在0轴下金叉');
EXPLAIN( 0轴上金叉 and not(0轴上二次金叉) and not(0轴上三次金叉), 'DEA和DIF在0轴上金叉');
EXPLAIN( 0轴上死叉, 'DEA和DIF在0轴上死叉');
EXPLAIN( 0轴上二次金叉, 'DEA和DIF在0轴上二次金叉');
EXPLAIN( 0轴上三次金叉, 'DEA和DIF在0轴上三次金叉');
EXPLAIN(DIF>=0 AND MACD>=0 AND MACD<=REF(MACD,1), '上涨趋势有所减缓, 短线投资者可考虑先逢高减持');
EXPLAIN(DIF<0 AND MACD>0 AND MACD>REF(MACD,1), '目前处于反弹阶段, 保持谨慎');
EXPLAIN(MACD>0 AND DIF>0 AND DIF>DEA AND DIF>REF(DIF,1) AND C>REF(C,1), '上升趋势尚未改变');
EXPLAIN(DIF<0 AND MACD>0 AND MACD< REF(MACD,1), '短线反弹已经结束, 应继续坚持谨慎原则');
EXPLAIN(DIF<0 AND MACD<=0 AND MACD>REF(MACD,1), '下跌趋势有所减缓, 可适当关注');
EXPLAIN(MACD<0 AND DIF<0 AND DIF< DEA AND DIF< REF(DIF,1) AND C< REF(C,1) AND MACD<=REF(MACD,1), '下降趋势尚未改变');
EXPLAIN(DIF>=0 AND MACD<0 AND MACD< REF(MACD,1), '目前正处于回落整理阶段');
EXPLAIN(DIF>=0 AND MACD<0 AND MACD>=REF(MACD,1), '目前仍处于整理阶段, 但整理即将结束, 可在均线或其他支撑位附近逢低介入');
{-----引用趋势顶底指标分析-----}
A:=(MA(-100*(HHV(HIGH,34)-CLOSE)/(HHV(HIGH,34)-LLV(LOW,34)),19),COLORRED);
B:=(-100*(HHV(HIGH,14)-CLOSE)/(HHV(HIGH,14)-LLV(LOW,14)));
d:=EMA(-100*(HHV(HIGH,34)-CLOSE)/(HHV(HIGH,34)-LLV(LOW,34)),4),linethick2;
长期线:=A+100; 短期线:=B+100; 中期线:=d+100;
见顶:=(ref(中期线,1)>85 and ref(短期线,1)>85 and ref(长期线,1)>65) and cross(长期线,短期线);
顶部区域:=(中期线<ref(中期线,1) and ref(中期线,1)>80) and (ref(短期线,1)>95 or ref(短期线,2)>95) and 长期线>60 and 短期线<83.5 and 短期线<中期线 and 短期线<长期线+4;
底部区域:=(长期线<12 and 中期线<8 and (短期线<7.2 or ref(短期线,1)<5) and (中期线>ref(中期线,1) or 短期线>ref(短期线,1)))
or (长期线<8 and 中期线<7 and 短期线<15 and 短期线>ref(短期线,1)) or (长期线<10 and 中期线<7 and 短期线<1);
EXPLAIN( 顶部区域, '走势有超买迹象, 需要引起注意'),COLORFFCC99;
EXPLAIN( 底部区域, '走势有超跌迹象, 需要引起注意'),color9966FF;
{=====【均线信息提示】=====}
EXPLAIN(C>0,'——【均线趋势分析】——'),COLORYELLOW;
一阴穿两线:=close/open<0.985 and high>MA1 and close<MA3;
一阳穿两线:=close/open>1.016 and low<min(MA1,MA3) and close>max(MA1,MA3);
均线密集:=上升趋势 and (max(MA1,MA4)/min(MA1,MA4))<1.01 and MA1>MA2;
均线发散程度:=MA1/MA4;
均线向下发散:=空头排列 and 均线发散程度<0.90 and 均线发散程度<0.99*ref(均线发散程度,1);
均线向上发散:=多头排列 and 均线发散程度>1.070 and 均线发散程度>(1.013*ref(均线发散程度,1));
上升趋势形成:=上升趋势 and MA1>ref(MA1,1) and ref(not(上升趋势),1);
EXPLAIN(上升趋势 and MA1>ref(MA1,1) and ref(not(上升趋势),1), '上升趋势形成'),colored;

```

```

EXPLAIN(上升趋势 and not(一阴穿两线) and ref(上升趋势,1),'处于上升趋势'),color9966FF;
EXPLAIN((not(上升趋势) or 一阴穿两线) and ref(上升趋势,1),'上升趋势停顿,注意行情可能转折'),color88CC66;
EXPLAIN(下跌趋势 and MA1<ref(MA1,1) and ref(not(下跌趋势),1),'下跌趋势形成');
EXPLAIN(下跌趋势 and not(一阳穿两线) and ref(下跌趋势,1),'处于下跌趋势');
EXPLAIN((not(下跌趋势) or 一阳穿两线) and ref(下跌趋势,1) and not(上升趋势形成),'下跌趋势趋缓,注意行情可能转折'),color00CCFF;
EXPLAIN(多头排列 and not(均线向上发散),'均线多头排列');
EXPLAIN(多头排列 and 均线向上发散,'均线多头向上发散,注意K线形态和股价运行节奏变化'),color3333CC;
EXPLAIN(空头排列 and not(均线向下发散),'均线空头排列');
EXPLAIN(空头排列 and 均线向下发散,'均线空头向下发散,注意K线形态和股价运行节奏变化'),color3333CC;
EXPLAIN(均线密集,'均线密集排列,注意走势方向变化'),color6633CC;
{=====K线特征及形态组合
=====}
涨停板:=CLOSE/REF(CLOSE,1)>1.097;
跌停板:={close<=open and} CLOSE/REF(CLOSE,1)<0.91;
K线实体:=abs(close-open);
阳线:=CLOSE>OPEN or 涨停板;
带下影阳线:=(阳线 or 涨停板) and (close-low)>1.6*(close-open);
光脚阳线:=阳线 and (open-low)<0.36*(close-open);
中阳线:=close/open>1.030;
短上影阳线:=阳线 and (high-close)<(high-open)*0.34;
阴线:=CLOSE<OPEN;
中阴线:=close/ref(close,1)<0.970;
平盘:=CLOSE=OPEN;
平台整理:=HHV(close,5)/LLV(open,5)<1.08;
{=====:
长十字:=CLOSE=OPEN AND high>open and HIGH/LOW >1.06;
垂死十字:=CLOSE=OPEN AND CLOSE=LOW AND CLOSE<>HIGH;
倒转锤头:=HIGH-LOW > 2.5*(MAX(OPEN,CLOSE)-LOW) AND CLOSE<min(MA1,MA2) and (min(open,close)-low)/(high-low)<0.10 and
open<MA1 and MA1<MA2 and open<>close;
锤头:=HIGH = MAX(OPEN,CLOSE) AND HIGH-LOW>3*(HIGH-MIN(OPEN,CLOSE)) AND CLOSE<MA(CLOSE,10);
吊颈:=(OPEN<>CLOSE and ((HIGH-MAX(OPEN,CLOSE))/(high-low))<0.16 ) AND (HIGH-LOW)>2.48*(abs(OPEN-CLOSE)) AND
CLOSE>MA1 and (high-low)/low>0.020;
十字星:=(abs(CLOSE-OPEN))/(high-Low)<0.19 and (abs(CLOSE-OPEN)/CLOSE)<0.009 AND HIGH>LOW and max(open,close)<high and
min(open,close)>low
and not(长十字) and not(倒转锤头) and not(吊颈);
早晨之星:=REF(CLOSE,2)/REF(OPEN,2)<0.97 AND REF(OPEN,1)<=REF(CLOSE,2) AND (REF(十字星,1) or ref(长十字,1)) AND CLOSE/OPEN>1.05
AND CLOSE>REF(CLOSE,2);
黄昏之星:=REF(CLOSE,2)/REF(OPEN,2)>1.03 AND REF(OPEN,1)>=REF(CLOSE,2) AND (REF(十字星,1) or ref(长十字,1)) AND CLOSE/OPEN<0.97
AND CLOSE<REF(CLOSE,2);
射击之星:=MIN(OPEN,CLOSE)=LOW AND HIGH-LOW > 3*(MAX(OPEN,CLOSE)-LOW) AND CLOSE > MA(CLOSE,5) and MA1>MA2;
升势尽头:=ref(阳线,1) and ref(high,1)/ref(close,1)>1.008 and (abs(close-open)/open<0.018 or 吊颈) and (max(close,open)<ref(high,1) or
(close<open
and max(close,open)<=ref(high,1))) and min(close,open)>ref(close,1) and MA1>MA2;
跌势尽头:=ref(阴线,1) and ref(阴线,2) and ref(close,1)/ref(open,1)<0.98 and ref(low,1)/ref(close,1)<0.988 and (abs(close-open)/open<0.025
or 倒转锤头) and max(close,open)<ref(close,1)
and min(close,open)>ref(low,1) and 空头排列;
穿头破脚1:=ABS((REF(CLOSE,1)-REF(OPEN,1))/REF(OPEN,1)<0.02 AND CLOSE/OPEN<0.986 AND CLOSE<min(REF(OPEN,1),REF(CLOSE,1))
AND OPEN>max(REF(CLOSE,1),REF(open,1))) and ref(Open,1)>max(MA1,MA2);
穿头破脚2:=(REF(CLOSE,1)/REF(OPEN,1)<0.990 AND CLOSE/OPEN>1.03 AND CLOSE>REF(OPEN,1) AND OPEN<REF(CLOSE,1));
乌云盖顶:=REF(CLOSE,1)/REF(OPEN,1)>1.02 AND CLOSE/OPEN<0.97 AND OPEN>REF(CLOSE,1) AND CLOSE<((REF(CLOSE,1) +
REF(OPEN,1))/2) and close>ref(open,1);
曙光初现:=REF(CLOSE,1)/REF(OPEN,1)<0.97 AND CLOSE/OPEN>1.04 AND OPEN<REF(CLOSE,1) AND close>(
REF(open,1)+REF(close,1))/2 and ref(close,1)<MIN(ref(MA1,1),ref(MA2,1));
旭日东升:=REF(CLOSE,1)/REF(OPEN,1)<0.985 AND 阳线 and close/open>1.020 AND OPEN>REF(CLOSE,1) AND OPEN<REF(open,1) AND
CLOSE>REF(OPEN,1) AND LOW<MA1 and ref(空头排列,1);
身怀六甲1:=ABS(REF(CLOSE,1)-REF(OPEN,1))/REF(CLOSE,1)>0.017 AND ABS(CLOSE-OPEN)/CLOSE<0.05 AND MAX(CLOSE,OPEN)
<MAX(REF(CLOSE,1),REF(OPEN,1)) AND MIN(CLOSE,OPEN) > MIN(REF(CLOSE,1),REF(OPEN,1))
and ref(阳线,1) and ref(close,1)>MA1;
身怀六甲2:=ABS(REF(CLOSE,1)-REF(OPEN,1))/REF(CLOSE,1)>0.017 AND ABS(CLOSE-OPEN)/CLOSE<0.04 AND MAX(CLOSE,OPEN)
<MAX(REF(CLOSE,1),REF(OPEN,1)) AND MIN(CLOSE,OPEN) > MIN(REF(CLOSE,1),REF(OPEN,1))
and ref(阴线,1) and ref(close,1)<MA2;

```

十字胎 :=ABS(REF(CLOSE,1)-REF(OPEN,1))/REF(CLOSE,1)>0.04 AND CLOSE=OPEN AND CLOSE < MAX(REF(CLOSE,1),REF(OPEN,1)) AND CLOSE > MIN(REF(CLOSE,1),REF(OPEN,1));
 平顶 :=ABS(HIGH-REF(HIGH,1))/HIGH<0.0005 and max(open,close)>max(MA2,MA4) and max(high,ref(high,1))>=hhv(high,8) and 阴线;
 平底 :=(ABS(LOW-REF(LOW,1))/LOW<0.001 and abs(ref(low,1)-ref(low,2))/ref(low,1)<=0.001) and 阳线 and min(open,close)<MA1;
 大阴烛 :=OPEN/CLOSE>1.06 AND HIGH/LOW <(OPEN/CLOSE+0.030);
 大阳烛 :=CLOSE/OPEN>1.07 AND 100*(HIGH/LOW) < 100*(CLOSE/OPEN)+2.3;
 光脚阴线 :=LOW=CLOSE and close/open<0.97;
 光脚长阴 :=光脚阴线 and 大阴烛;
 光头阳线 :=HIGH=CLOSE AND HIGH<>LOW and 中阳线;
 低开大阳 :=CLOSE/OPEN>1.06 AND open=low and Open<ref(low,1) and open<MA1;
 双飞乌鸦 :=REF(阴线,1) AND 阴线 AND ref(阳线,2) AND CLOSE/OPEN<0.988 and close<ref(close,1) and ref(not(阴线),2) and ref(多头排列,2) and open<=ref(close,1);
 多方炮 :=REF(阳线,2) and ref(close,2)/ref(open,2)>1.014 AND REF(阴线,1) AND 阳线 and close>(ref(close,1)+ref(open,1))/2 and close/open>1.023 and open<ref(open,1);
 空方炮 :=REF(阴线,2) and ref(open,2)/ref(close,2)>1.02 AND REF(阳线,1) AND 阴线 and close<(ref(close,1)+ref(open,1))/2 and open/close>1.03 and low<ref(open,1) ;
 出水芙蓉 :=(OPEN<EMA(C,10) OR OPEN<EMA(C,20)) AND (C>MAX(EMA(C,10),MAX(EMA(C,20),EMA(C,30)))) AND (V/MA(V,30)>1.2 AND C/REF(C,1)>1.049);
 上升三部曲 :=REF(CLOSE,4)/REF(OPEN,4)>1.016 AND REF(阴线,3) AND REF(阴线,2) AND REF(close<=open,1) AND REF(LOW,4)<REF(LOW,3) AND REF(LOW,4)<REF(LOW,2) AND REF(LOW,4)<REF(LOW,1) AND CLOSE/OPEN>1.015 AND CLOSE>REF(CLOSE,4) AND REF(HIGH,4)>REF(close,3) AND REF(HIGH,4)>REF(close,2) AND REF(HIGH,4)>REF(close,1) and (HIGH-CLOSE)/(HIGH-OPEN)<0.30;
 上升二法 :=REF(CLOSE,3)/REF(OPEN,3)>1.02 AND REF(阴线,2) AND REF(close<=open,1) AND ref(open,1)<ref(close,2) AND REF(LOW,3)<REF(LOW,2) AND REF(LOW,3)<REF(LOW,1) AND CLOSE/OPEN>1.013 AND CLOSE>REF(CLOSE,3) ;
 下跌三部曲 := REF(中阴线,4) AND REF(阳线,3) AND REF(阳线,2) AND REF(阳线,1) AND REF(close,4)<REF(open,3) AND REF(close,4)<REF(open,2) AND REF(close,4)<REF(open,1) AND REF(HIGH,4)>REF(HIGH,3) AND REF(HIGH,4)>REF(HIGH,2) AND REF(HIGH,4)>REF(close,1) AND 中阴线 AND CLOSE<REF(CLOSE,4);
 跳空缺口1 :=LOW>REF(HIGH,1) and ref(not(多头排列),1);
 跳空缺口2 :=HIGH<REF(LOW,1);
 见顶缺口 :=LOW>REF(HIGH,1) and open>MA1 and (high-max(close,open))/close>0.02 and 多头排列 and 均线向上发散 and not(均线密集) and ref(close,1)>UPPER;
 三只乌鸦 :=REF(阴线,2) AND REF(阴线,1) AND 阴线 AND REF(CLOSE,1)<REF(CLOSE,2) AND CLOSE<REF(CLOSE,1) and ref(close>open ,3) and REF(open,2)>max(ref(MA1,2),ref(MA2,2)) and ref(多头排列,3) and close<MA2;
 加速三阴 :=(REF(阴线,2) or (ref(平盘,2) and ref(open,2)<ref(MA1,2))) AND REF(阴线,1) AND 阴线 AND REF(CLOSE,1)<REF(CLOSE,2)*0.95 AND CLOSE<REF(CLOSE,1) and REF(close,2)<min(ref(MA1,2),ref(MA3,2)) and ref(空头排列,1) and close/open<0.975 and (close-low)/(open-low)<0.35 and ((open<= (ref(close,1)*1.005) and close/open<0.95) or (open<ref(close,1)));
 冲顶三阳 :=(REF(阳线,2) AND REF(阳线,1) AND 阳线) and (K线实体/open)>0.04 and K线实体>ref(K线实体,1)*0.9 and ref(K线实体,1)>ref(K线实体,2) and 多头排列 and ref(CLOSE,1)/ref(close,2)>1.030 and open>=ref(close,1) and (not(涨停板) or (涨停板 and 换手率>20));
 看涨分离 :=OPEN=REF(OPEN,1) AND (CLOSE-OPEN)*(REF(CLOSE,1)-REF(OPEN,1))<0 and ref(阴线,1) and not(十字星);
 看跌分离 :=OPEN=REF(OPEN,1) AND (CLOSE-OPEN)*(REF(CLOSE,1)-REF(OPEN,1))<0 and ref(阳线,1);
 长下影 :=(MIN(CLOSE,OPEN)-LOW)/(HIGH-LOW)>0.667 and (min(close,open)-Low)/Low>0.02 and not(涨停板);
 长上影 :=(HIGH-MAX(CLOSE,OPEN))/(HIGH-LOW)>0.38 and (high-max(close,open))/close>0.025 and max(open,close)>MA1 and not(倒转锤头);
 好友反攻 :=(REF(阴线,1) AND 阳线 AND ABS(CLOSE-REF(CLOSE,1))/CLOSE<0.002) and 中阳线;
 坏友反攻 :=(REF(阳线,1) AND 阴线 AND ABS(CLOSE-REF(CLOSE,1))/CLOSE<0.002) and 中阴线;
 看涨吞没 :=ref(阴线,1) and 中阳线 and open<=min(ref(open,1),ref(close,1)) and close>max(ref(open,1),ref(close,1)) and not(穿头破脚2) and (HIGH-CLOSE)/(HIGH-OPEN)<0.30;
 看跌吞没 :=ref(阳线,1) and 中阴线 and open>=max(ref(open,1),ref(close,1)) and close<min(ref(open,1),ref(close,1)) and not(穿头破脚1);
 倾盆大雨 :=REF(C,1)/REF(O,1)>=1.02 AND O<REF(C,1) AND O>REF(O,1) AND C<REF(O,1) and close/open<0.965 and ref(多头排列,1);
 断头铡刀 :=close/open<0.98 and ref(MA1,1)>ref(MA4,1) and close<min(MA2,MA4) and open>=max(MA1,MA2) and (close-low)/low<0.015 and close<min(ref(open,1),ref(close,1));
 {=====